

Safety Management System

Smart. Manageable. Simple.

AEA Has Your Solution for SMS



Welcome

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SMS Training

- This is Coordinator Training.

- May be used as recurrent training.

SMS is an
“evolutionary” change
not a
revolutionary change.

Agenda

- Part 1: Introduction to Safety Management Systems
- Part 2: Our (AEA) Approach
- Part 3: SMS Elements
- Part 4: AEA's Online SMS Management Tool
- Part 5: Getting Started

Part 1:

**Introduction to
Safety Management Systems**

Definitions:

- Goals & Objective: WHAT we want to achieve.
- Management System: HOW we achieve our goals.

As A Business

- What are your expectations?

Objectives

- What are your objectives?

Safety Goal

- A certificated repair station must establish and maintain a quality control system acceptable to the FAA that ensures the airworthiness of the articles on which the repair station or any of its contractors performs work.

■ (14 CFR § 145.211 (a))

Safety Goal

- Every take off has a successful landing
- Every employee who walks into work – walks away from work.
- Every part works as advertised – every time

Safety Objectives

- Impediments to achieving your GOAL.
- What keeps you from being perfect?
- Established and reviewed annually

Is it working?

- How do you measure success?
- What incremental objective(s) DO YOU need to support your overall safety goal?

HOW you will achieve your goals ...

Your Management System for Safety.

SMS

SMS is a tool

...to help you achieve YOUR objectives!!



What SMS is NOT!



It is not a
package

Safety Management System

- SMS is not a “safety objective”
- SMS is a tool to help you achieve your safety objective.

What is Safety?

- "The condition of being safe from undergoing or causing hurt, injury, or loss"

(merriam-webster)

In classic terms...

Safety is the absence of risk...

From ICAO:

- While the elimination of accidents (and serious incidents) would be desirable, a one hundred per cent safety rate is an unachievable goal.
- Failures and errors will occur, in spite of the best efforts to avoid them.
- No human activity or human-made system can be guaranteed to be absolutely safe, i.e. free from risk.

From ICAO:

Safety is a relative notion whereby inherent risks are acceptable in a “safe” system.

From ICAO:

Safety is increasingly viewed as the management of risk.

What is Safety?

Safety is the state in which the risk of harm to persons or of property damage is reduced to, and maintained at or below, an acceptable level through a continuing process of hazard identification and risk management.

(ICAO)

Safety is:

- The management of Risk:
 - Past
 - Present
 - Future

Safety Management Systems

- SMS is simply the “Management System” used to achieve your desired “safety” goals and objective.

Safety Management Systems

WHY?

Safety Management Systems

The Secretary of Transportation shall assign and maintain safety as the highest priority in air commerce.

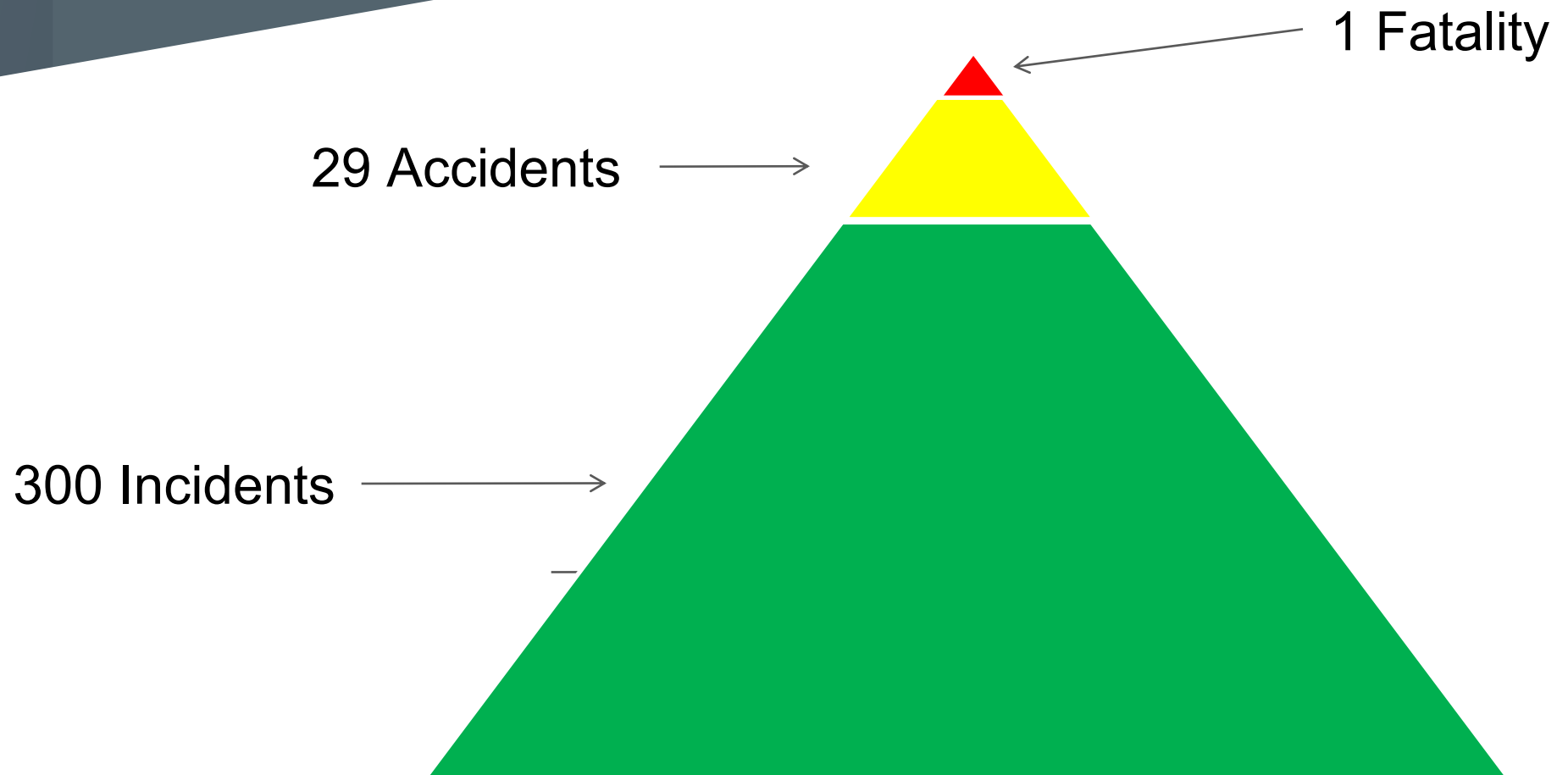
(TITLE 49 – TRANSPORTATION)



The Evolution of Safety

- The Technical Era:
(early 1900s until the late 1960s)
- The Human Factors Era
(early 1970s until the mid-1990s)
- The Organizational Era —
(mid-1990s to the present day)

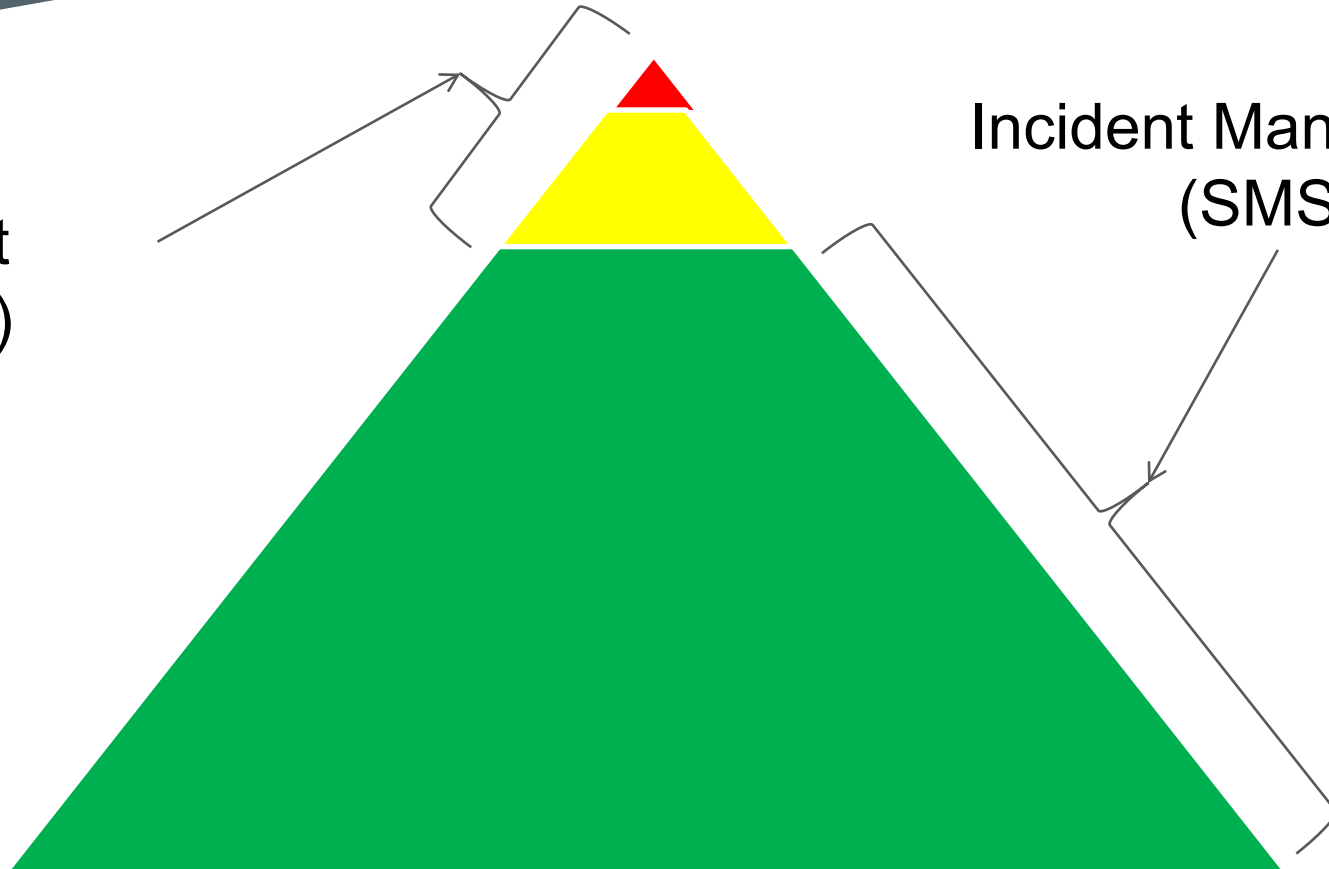
Accident Pyramid



Why SMS?

Accident
Management
(Rulemaking)

Incident Management
(SMS)



Why SMS?

- Manage the precursors to an accident or incident.

The Business Side:

■ Accidents are BAD business:

■ To the Customer

■ To the Employer

■ To the Employee

Costs of Accidents

- There are two basic types of costs associated with an accident or a serious incident: direct and indirect.

Direct Costs

- These are the obvious costs which are fairly easy to determine.
- They mostly relate to physical damage and include rectifying, replacing or compensating for injuries, aircraft equipment and property damage.
- The high costs of an accident can be reduced by insurance coverage.

Indirect Costs

- Indirect costs include all those items that are not directly covered by insurance.
- Uninsured costs typically include:
 - a) *Loss of business and damage to the reputation of the organization.*
 - b) *Loss of use of equipment.*
 - c) *Loss of staff productivity.*
 - d) *Investigation and clean-up.*
 - e) *Insurance deductibles*
 - f) *Legal action and damage claims.*
 - g) *Fines and citations.*

Why SMS?

- To manage and minimize the cost of accidents or incidents.

Why SMS?

- Laws
- Regulations
- ICAO
- Culture
- Public Expectation

Core Business Function

- Safety management is a core business function.
- Experience in other industries emphasize the importance of managing safety in a systematic, proactive and explicit manner.

Why SMS?

- Safety management is a core business function.

WHAT are your goals ...

- Business?
- Financial?
- Workforce?
- Production?
- Quality?

HOW will you achieve your goals ...

- Business Management System.
- Financial Management System.
- Workforce Management System.
- Production Management System.
- Quality Management System.

Is it working?

■ What “tools” do you use to measure success?

- Business.
- Financial.
- Workforce.
- Production.
- Quality.

What is a Management System?

- Management system refers to what the organization does to manage its processes, or activities, so that its products or services meet the objectives it has set itself, such as:
 - satisfying the **customer's quality requirements**,
 - complying with **regulations**, or
 - meeting **safety objectives**.

Quality Management System (QMS)

- QMS: The *Management System* to support the chosen level of quality
 - Quality Control
 - Quality Assurance

Safety Management Systems

- SMS is simply the “Management System” used to achieve the desired “safety” objective.

Safety Management Systems

- What is the “Safety Objective”?

- “The limiting of accident risk through design of a systematic, methodical, and forward-looking hazard identification and control program, beginning in the conceptual design phase and continuing through its development, fabrication, testing, production, use, and ultimate disposal.”

System Safety

- “The limiting of accident risk through design of a systematic, methodical, and forward-looking hazard identification and control program, beginning in the conceptual design phase and continuing through its development, fabrication, testing, production, use, and ultimate disposal.”

Safety Management Systems

- What is the “Safety Objective”?

SYSTEM SAFETY

Safety Management Systems

- So...What is SMS?
- The *Management System* designed to manage and promote *System Safety*.

Systematic

- means that safety management activities will be conducted in accordance with a predetermined plan and applied in a consistent manner throughout the organization.

Proactive

- means the adoption of an approach which emphasizes prevention through the identification of hazards and the introduction of risk mitigation measures before the risk-bearing event occurs and adversely affects safety performance.

Explicit

- means that all safety management activities should be documented, visible and performed independently from other management activities.

Strategies for Safety Management

- Old School: Reactive safety strategy: Investigate accidents and reportable incidents
- Today: Proactive safety strategy: Aggressively seeking information from a variety of sources which may be indicative of emerging safety problems

Typical Hazards in the workplace

Tripping Hazard



ICAO SMS principles

- Safety Policy
- Safety Risk Management
- Safety Assurance
- Safety Promotion

Safety Risk Management

■ Safety risk management includes the following:

- a) a definition of safety risk;
- b) safety risk probability;
- c) safety risk severity;
- d) safety risk tolerability; and
- e) safety risk management.

Definition of safety risk

- Safety risk is the projected likelihood and severity of the consequence or outcome from an existing hazard or situation.
- While the outcome may be an accident, an “intermediate unsafe event/consequence” may be identified as “the most credible outcome”.

Evaluating Risk

- Likelihood of occurrence of the hazard;
and,
- The severity of its potential
consequences.

Perceptions of Risk

- Three broad categories:
 - a) risks that are so high that they are unacceptable;
 - b) risks that are so low that they are acceptable; and
 - c) risks in between the two categories in a) and b), where consideration needs to be given to the various trade-offs between risks and benefits.

Risk Acceptability

- If the risk does not meet the predetermined acceptability criteria, an attempt must always be made to reduce it to a level that is acceptable, using appropriate mitigation procedures.

Risk Acceptability

- If the risk cannot be reduced to or below the acceptable level, it may be regarded as tolerable if:
 - a) the risk is less than the predetermined unacceptable limit;
 - b) the risk has been reduced to a level that is as low as reasonably practicable; and
 - c) the benefits of the proposed system or changes are sufficient to justify accepting the risk.

- *Note. — All three of the above criteria should be satisfied before a risk is classed as tolerable*

Accidents Verses Incidents

- An accident is an occurrence during the operation of an aircraft which entails:
 - 1) a fatality or serious injury;
 - 2) substantial damage to the aircraft involving structural failure or requiring major repair; or
 - 3) the aircraft is missing or is completely inaccessible.

Accidents Verses Incidents

- An incident is an occurrence, other than an accident, associated with the operation of an aircraft which affects or could affect the safety of operation.
- A serious incident is an incident involving circumstances indicating that an accident nearly occurred.

Accident Causation

- The strongest evidence of a serious breach of a system's safety is an accident.

Traditional view of causation

- Following a major accident, we ask:
 - a) *How and why did competent personnel make the errors necessary to precipitate the accident?*
 - b) Could something like this happen again?

Accident Causation Model

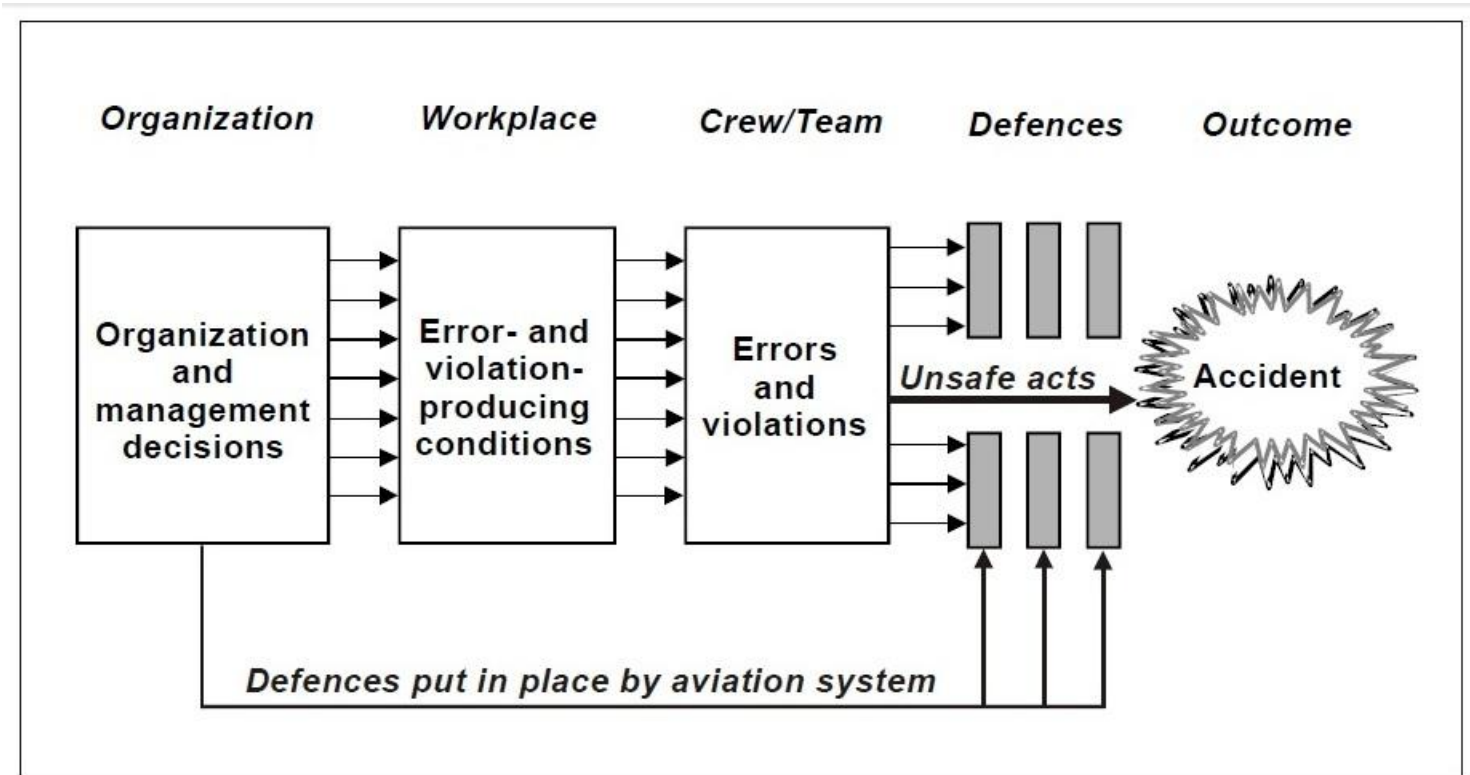


Figure 4-2. Accident causation model
(Adapted from Prof. James Reason)

Modern View of Causation

- Accidents require the coming together of a number of enabling factors — each one necessary but in itself not sufficient to breach system defenses.

Modern View of Causation

- Major equipment failures or operational personnel errors are seldom the sole cause of breaches in safety defenses.

Modern View of Causation

- Often breakdowns are the consequence of human failures in *decision-making*.
- The breakdowns may involve *active failures* at the operational level, or *latent conditions* conducive to facilitating a breach of the system's inherent safety defenses.
- Most accidents include both active and latent conditions.

Modern View of Causation

- *Active failures:*
 - Errors and violations having an immediate adverse effect.
 - These unsafe acts may be the result of normal errors, or they may result from deliberate violations of prescribed procedures and practices.

Modern View of Causation

- *Latent conditions:*
 - A latent condition is the result of an action or decision made well before an accident.
 - Its consequences may remain dormant for a long time.
 - Individually, these latent conditions are usually not harmful since they are not perceived as being failures in the first place.

Context for Accidents and Incidents

- Accidents and incidents occur within a defined set of circumstances and conditions. These include:
 - the aircraft and other equipment,
 - the weather,
 - the airport and flight services,
 - the operating climate (regulatory, industry and corporate).
 - human behavior.

Part 2:

Our Approach

Our Approach

- Start with ICAO Annex 19
- Utilizing ICAO Safety Management Manual (SMM)
- Perform a gap analysis of the ICAO requirements and standard part 145 (AMO/Repair Station) regulations
- Develop SMS management tool

Our Approach

- Focus on:
 - The regulatory, industry and corporate operating climate.
 - Human factors.
 - Quality assurance
 - Emergency Response

Gap Analysis

- A gap analysis compares the service provider's existing safety management processes and procedures with requirements contained in the SMS framework.

Gap Analysis

- Aviation service providers will have typically implemented various SMS functions due to their compliance with national regulations or adoption of industry best practices.
- The development of an SMS should *build upon* existing organizational structures and control systems

Our Approach

- The relationship between SMS and QMS leads to the complementary contributions of each system to the attainment of the organization's safety and quality goals.

Our Approach

Summary comparison of QMS and SMS

<i>QMS</i>	<i>SMS</i>
Quality	Safety
Quality assurance	Safety assurance
Quality control	Hazard identification and risk control
Quality culture	Safety culture
Compliance with requirements	Acceptable level of safety performance
Prescriptive	Performance-based
Standards and specifications	Organizational and human factors
Reactive > Proactive	Proactive > Predictive

§ 145.211 (a)

- A certificated repair station must establish and maintain a quality control system acceptable to the FAA that ensures the airworthiness of the articles on which the repair station or any of its contractors performs maintenance, preventive maintenance, or alterations.

We Ask

- How did a *competent person* make the errors necessary to contribute/cause the accident?
- Why did the *competent person* make the errors necessary to cause the accident?
- Could something like this happen again?

The Regulations Address Competency

- Each certificated repair station provide qualified personnel to plan, supervise, perform, and approve for return to service the maintenance, preventive maintenance, or alterations performed under the repair station certificate and operations specifications;

§ 145.151 (b)

The Regulations Address Competency

- The training program must ensure each employee assigned to perform maintenance, preventive maintenance, or alterations, and inspection functions is capable of performing the assigned task.

§ 145.163 (b)

Why did the *competent person* make the errors necessary to cause the accident?

Human Factors & Organizational Factors

Human Factors

The PEAR Model

- The PEAR model is a simple framework for thinking about human factors in aviation maintenance.
- PEAR is an acronym for People, Environment, Actions, and Resources.

Components of the PEAR model.

- PEOPLE - Who are:

- Physiological Factors

- Lifestyle

- Fatigue

- Drugs

- Physical limitations

Components of the PEAR model.

- PEOPLE - Who are:

- Psychological Factors

- Workload
 - Experience
 - Knowledge
 - Attitude
 - Mental or emotional state

§ 145.151 Personnel requirements.

- Each certificated repair station must--
 - (c) Ensure it has a sufficient number of employees with the training or knowledge and experience in the performance of maintenance, preventive maintenance, or alterations authorized by the repair station certificate and operations specifications to ensure all work is performed in accordance with part 43; and

Components of the PEAR model.

- PEOPLE - Who are:

- Psychosocial Factors

- Interpersonal conflicts

- Personal loss

- Financial hardships

Components of the PEAR model.

- ACTIONS - What do we do?

- Communication

- Oral

- Visual

- Written

- Information and Control Requirements

Inspection and Quality Control System

- Continuity of Inspection Responsibility. Include procedures for ensuring that the responsibilities of its inspectors are properly performed if they cannot complete the task.
- If the facility uses multiple shifts, include procedures to ensure a continuing responsibility for maintenance in progress with status books, shift change logs, or similar documents.

Components of the PEAR model.

- ENVIRONMENT - Where do we work?

- Physical

- Weather
 - Location inside/outside
 - Workspace
 - Lighting
 - Noise
 - Safety

§ 145.103 Housing and facilities requirements.

- (a) Each certificated repair station must provide—
 - (2) Facilities for properly performing the maintenance, preventive maintenance, or alterations of articles or the specialized services for which it is rated.

§ 145.103 Housing and facilities requirements.

- (a)(2) ...Facilities must include the following:
 - (v) Ventilation, lighting, and control of temperature, humidity, and other climatic conditions sufficient to ensure personnel perform maintenance, preventive maintenance, or alterations to the standards required by this part.

Components of the PEAR model.

- ENVIRONMENT - Where do we work?
 - Organizational
 - Personnel
 - Supervision
 - Shift
 - Pressures
 - Crew structure

Components of the PEAR model.

- RESOURCES –

- What and whom do we use to do our job?

- Work Cards
 - Manuals/Bulletins/FARs
 - Test Equipment
 - Computers
 - Paperwork/Signoffs

§ 145.109 Equipment, materials, and data requirements

- (d) A certificated repair station must maintain, in a format acceptable to the FAA, the documents and data required for the performance of maintenance, preventive maintenance, or alterations under its repair station certificate and operations specifications in accordance with part 43.

Components of the PEAR model.

- RESOURCES –

- What and whom do we use to do our job?

- Ground Handling Equipment

- Forklifts/Tow Motors

- Ladders/Steps/Work Platforms

- Cranes Hoist/Jacks

Components of the PEAR model.

■ RESOURCES –

■ What and whom do we use to do our job?

■ Fixtures

■ Materials

■ Task Lighting

■ Training

Organizational Factors

System Safety

- “The limiting of accident risk through design of a systematic, methodical, and forward-looking hazard identification and control program, beginning in the conceptual design phase and continuing through its development, fabrication, testing, production, use, and ultimate disposal.”

Corporate Safety Culture

- Organizational or corporate culture provides a cornerstone for managerial and employee decision-making —

“This is how we do things here!”

Corporate Safety Culture

- Safety culture is affected by such factors as:
 - a) management's actions and priorities;
 - b) policies and procedures;
 - c) supervisory practices;
 - d) safety planning and goals;
 - e) actions in response to unsafe behaviors;
 - f) employee training and motivation; and
 - g) employee involvement or “buy-in”.

SMS

■ The Management System that provides for

■ Safety Policy

■ Safety Promotion

AND

■ Safety Risk Management

■ Safety Assurance

Management's Actions and Priorities

- Safety first!
- Good business practices:
 - Protect the customer
 - Protect the customer's property
 - Protect the employees
 - Protect your environment
 - Make thoughtful, thorough decisions

Policies and Procedures

- Compliant
- Good business practices:
 - Comply with Federal Aviation Regulations
 - Comply with OSHA
 - Comply with EPA

Supervisory Practices

- Communication
- Good business practices:
 - Clear communication regarding:
 - Schedules
 - Mission
 - Acceptable Risk

Safety Planning and Goals

- Safety Goal: Zero accidents/incidents
- Good business practices:
 - Plan don't "wish".
 - Listen
 - See what you don't see
 - Pay attention to the little things.
 - Investigate incidents

Actions in Response to Unsafe Behaviors

- Appropriate
- Good business practices:
 - Manage behavior
 - Train/Educate
 - Appropriate Accountability
 - Encourage self reporting with non-punitive actions.

Employee Training and Motivation

- Train to expectations
- Good business practices:
 - Set the standard – train to the standard
 - Allow time to train
 - Two-way communications – see results
 - Reward positive actions

Employee Involvement or “buy-in”

- Without participation you only have a book!
- Good business practices:
 - Employee developed
 - Employee benefit
 - Easy/quick to use
 - Observable benefit
 - Just culture

Positive Safety Culture

- Informed cultures.
- Learning cultures.
- Reporting cultures.
- Just cultures.

Informed Cultures

- Management fosters a culture where people understand the hazards and risks inherent in their areas of operation.

Informed Cultures

- Personnel are provided with the necessary knowledge, skills and job experience to work safely, and they are encouraged to identify the threats to their safety and to seek the changes necessary to overcome them.

Learning Cultures

- Learning is seen as more than a requirement for initial skills training; rather it is valued as a lifetime process.
- People are encouraged to develop and apply their own skills and knowledge to enhance organizational safety.

Learning Cultures

- Staff are updated on safety issues by management, and safety reports are fed back to staff so that everyone can learn the pertinent safety lessons.

Reporting Cultures

- Managers and operational personnel freely share critical safety information without the threat of punitive action.

Reporting Cultures

- Personnel are able to report hazards or safety concerns as they become aware of them, without fear of sanction or embarrassment.

Just Cultures

- While a non-punitive environment is fundamental for a good reporting culture, the workforce must know and agree on what is acceptable and what is unacceptable behavior.

Just Cultures

- Negligence or deliberate violations must not be tolerated by management (even in a non-punitive environment).

Just Cultures

- A just culture recognizes that, in certain circumstances, there may be a need for punitive action and attempts to define the line between acceptable and unacceptable actions or activities.

Characteristics of different safety cultures

Table 4-1. Characteristics of different safety cultures

Safety Culture: → Characteristics ↓	<i>Poor</i>	<i>Bureaucratic</i>	<i>Positive</i>
Hazard information is:	Suppressed	Ignored	Actively sought
Safety messengers are:	Discouraged or punished	Tolerated	Trained and encouraged
Responsibility for safety is:	Avoided	Fragmented	Shared
Dissemination of safety information is:	Discouraged	Allowed but discouraged	Rewarded
Failures lead to:	Cover-ups	Local fixes	Inquiries and systemic reform
New ideas are:	Crushed	Considered as new problems (not opportunities)	Welcomed

Part 3:

SMS Elements

ICAO SMS Framework

- The four components that comprise the ICAO SMS framework are:
 - Safety policy and objectives
 - Safety risk management
 - Safety assurance
 - Safety promotion

Safety Policy and Objectives

1. Management commitment and responsibility
2. Safety accountabilities
3. Appointment of key safety personnel
4. Coordination of emergency response planning
5. SMS documentation

Safety Risk Management

1. Hazard identification
2. Safety risk assessment and mitigation

Safety Assurance

1. Safety performance monitoring and measurement
2. The management of change
3. Continuous improvement of the SMS

Safety Promotion

1. Training and education
2. Safety communication

Safety Policy & Objectives

1. Management commitment and responsibility
2. Safety accountabilities
3. Appointment of key safety personnel
4. Coordination of emergency response planning
5. SMS documentation

Management Commitment

- The service provider shall define its safety policy in accordance with international and national requirements

Management Commitment

The safety policy shall:

- reflect organizational commitment regarding safety;
- include a clear statement about the provision of the necessary resources for the implementation of the safety policy;
- include safety reporting procedures;
- clearly indicate which types of behaviors are unacceptable related to the service provider's aviation activities and include the circumstances under which disciplinary action would not apply;
- be signed by the accountable executive of the organization;
- be communicated, with visible endorsement, throughout the organization; and
- be periodically reviewed to ensure it remains relevant and appropriate to the service provider.

Safety Accountabilities

The service provider shall:

- identify the accountable executive who, irrespective of other functions, has ultimate responsibility and accountability, on behalf of the organization, for the implementation and maintenance of the SMS;
- clearly define lines of safety accountability throughout the organization, including a direct accountability for safety on the part of senior management;

Safety Accountabilities

The service provider shall:

- identify the accountabilities of all members of management, irrespective of other functions, as well as of employees, with respect to the safety performance of the SMS;
- document and communicate safety responsibilities, accountabilities and authorities throughout the organization; and
- define the levels of management with authority to make decisions regarding safety risk tolerability.

Appointment of Key Personnel

- The service provider shall appoint a safety manager who is responsible for the implementation and maintenance of an effective SMS.

Emergency Response Planning

- The service provider shall ensure that an emergency response plan is properly coordinated with the emergency response plans of those organizations it must interface with during the provision of its services.

SMS Documentation

- The service provider shall develop an SMS implementation plan, formally endorsed by the organization, that defines the organization's approach to the management of safety in a manner that meets the organization's safety objectives.

SMS Documentation

- The service provider shall develop and maintain SMS documentation that describes:
 - the safety policy and objectives;
 - SMS requirements;
 - SMS processes and procedures;
 - accountabilities, responsibilities and authorities for SMS processes and procedures; and
 - SMS outputs.

SMS Documentation

- The service provider shall develop and maintain an SMS manual as part of its SMS documentation.

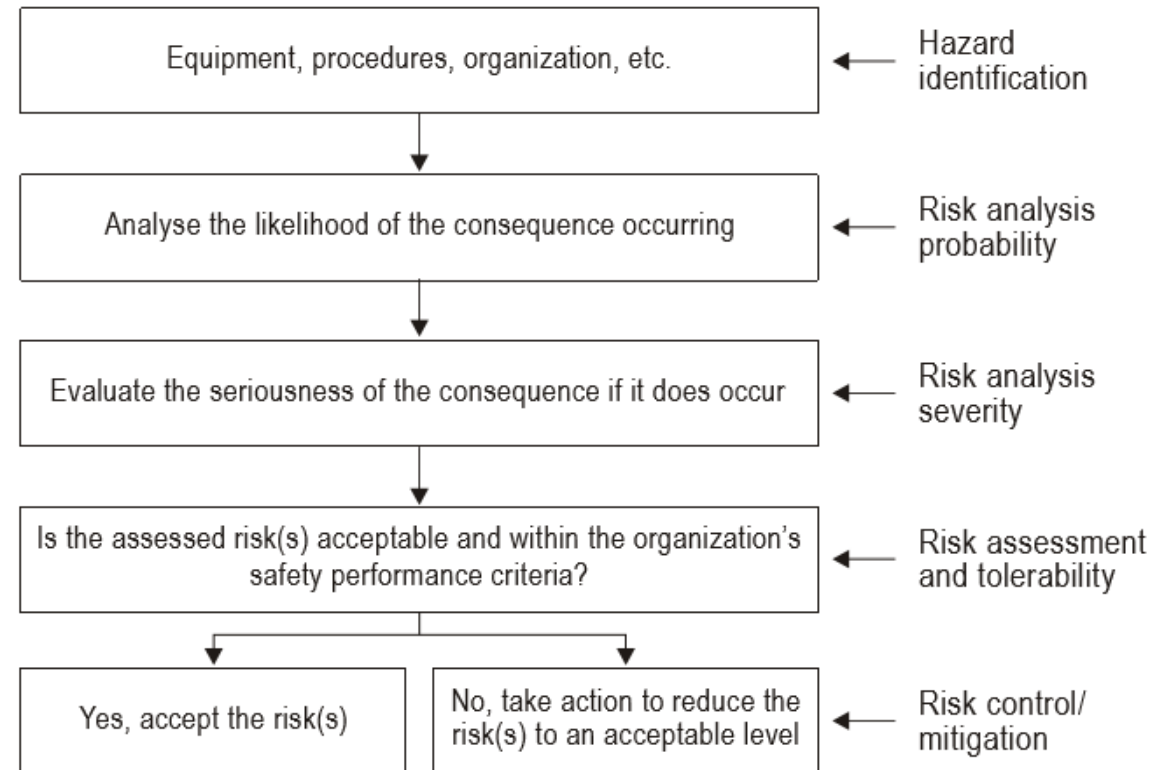
Safety Risk Management

1. Hazard identification
2. Safety risk assessment and mitigation

Hazard Identification

- The service provider shall develop and maintain a formal process that ensures that hazards associated with its aviation products or services are identified.
- Hazard identification shall be based on a combination of reactive, proactive and predictive methods of safety data collection.

Safety Risk Management Process



Risk Probability & Severity Table

Risk Probability	Risk Severity				
	Catastrophic A	Hazardous B	Major C	Minor D	Negligible E
Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E
Extremely Improbable 1	1A	1B	1C	1D	1E

Safety Risk Assessment & Mitigation

- The service provider shall develop and maintain a process that ensures analysis, assessment and control of the safety risks associated with identified hazards.

Effective Safety Reporting

- There are five basic characteristics that are universally associated with effective safety reporting systems.
- Effective hazard reporting is a key component of safety management. Once reported, data on hazards can be analyzed with other data sources to support the SRM and SA processes

Five Characteristics of Effective Reporting

1. Information

People are knowledgeable about the human, technical and organizational factors that determine the safety of the system as a whole.

Five Characteristics of Effective Reporting

2. Flexibility

People can adapt their reporting mode when facing unusual circumstances, shifting from the established mode to a direct mode thus allowing information to quickly reach the appropriate decision-making level.

Five Characteristics of Effective Reporting

3. Willingness

People are willing to report their errors and experiences

Five Characteristics of Effective Reporting

4. Learning

People have the competence to draw conclusions from safety information systems and the will to implement major reforms

Five Characteristics of Effective Reporting

5. Accountability

People are encouraged (and rewarded) for providing essential safety-related information.

However, there is a clear line that differentiates between acceptable and unacceptable behavior.

Safety Assurance

1. Safety performance monitoring and measurement
2. The management of change
3. Continuous improvement of the SMS

Safety Performance Monitoring & Measurement

- The service provider shall develop and maintain the means to verify the safety performance of the organization and to validate the effectiveness of safety risk controls.
- The service provider's safety performance shall be verified in reference to the safety performance indicators and safety performance targets of the SMS.

Safety Performance Indicators

- An SMS defines measurable performance outcomes to determine whether the system is truly operating in accordance with design expectations and not simply meeting regulatory requirements.
- The safety performance indicators are used to monitor known safety risks, detect emerging safety risks and to determine any necessary corrective actions.

Safety Performance Indicators

- A service provider's safety achievement as defined by its safety performance targets and safety performance indicators.

Safety Performance Indicators

- A data-based safety parameter used for monitoring and assessing safety performance.

Safety Performance Indicators

- The safety performance indicators and associated targets should be accepted by the State responsible for the service provider's authorization, certification or designation.
- Safety performance indicators are *supplementary* to any legal or regulatory requirements and do not relieve service providers from their regulatory obligations.

The Management of Change

- The service provider shall develop and maintain a formal process to identify changes which may affect the level of safety risk associated with its aviation products or services and to identify and manage the safety risks that may arise from those changes.

Change Management

- A formal process to manage changes within an organization in a systematic manner, so that changes which may impact identified hazards and risk mitigation strategies are accounted for, before the implementation of such changes.

Continuous Improvement of SMS

- The service provider shall monitor and assess the effectiveness of its SMS processes to enable continuous improvement of the overall performance of the SMS.

Safety Promotion

1. Training and education
2. Safety communication

Training & Education

- The service provider shall develop and maintain a safety training program that ensures that personnel are trained and competent to perform their SMS duties.
- The scope of the safety training program *shall be appropriate* to each individual's involvement in the SMS.

Safety Communication

The service provider shall develop and maintain formal means for safety communication that:

- a) ensures personnel are aware of the SMS to a degree commensurate with their positions;
- b) conveys safety-critical information;
- c) explains why particular safety actions are taken; and
- d) explains why safety procedures are introduced or changed.

Part 4

AEA's Online SMS Management Tool

S
M
S

How do we do this?



Safety Management Systems

Smart
Manageable
Simple

Hazard identification and risk management

- § 145.1 This part also contains the rules a certificated repair station must follow related to its performance of maintenance,

Hazard identification and risk management

- Advisory circular (AC) 145-9 provides information to develop and evaluate a Repair Station Manual (RSM) and Quality Control Manual (QCM)

(processes)

Hazard identification and risk management

- SMS provides guidance on identifying potential hazards and guides the applicant to mitigate those hazards to the maximum extent practical while defining the *business processes* (i.e describing the repair station organization and processes as required in part 145).

Basic Elements of SMS

- 1. Management commitment to the company Safety System
- 2. The SMS Coordinator (Safety Manager) is either the Accountable Manager or reports directly to the Accountable Manager

Basic Elements of SMS

- 3. Encouragement of a positive safety culture
- 4. Establishment of a safety management structure,

Basic Elements of SMS

- 5. Hazard identification and risk management,
- 6. Safety audits and assessment of quality and compliance,

Basic Elements of SMS

- 7. Accident and incident reporting and investigation,
- 8. Emergency Response Program Management,

Basic Elements of SMS

- 9. Documentation and Recordkeeping,
- 10. Immunity-based reporting systems,
- 11. The exchange of valuable “Lessons Learned” with others in the industry,

Basic Elements of SMS

- 12. Safety training integration into the organization's training syllabi,
- 13. Human factors training for all personnel and
- 14. Regular evaluation and ongoing fine-tuning of the Safety Management System.

AEA's Solution:

- All of the basic requirements
 - Regulatory compliance
 - Incident management
 - Positive Safety Culture

Informed Cultures

- People:
 - understand the hazards and risks inherent in their areas of operation.
 - are provided with the necessary knowledge, skills and job experience to work safely.
 - are encouraged to identify the threats to their safety and to seek the changes necessary to overcome them.

Indoctrination Training

Includes hazard identification, risk assessment and overview of the SMS program and participation.



Aircraft Electronics Association Indoctrination Training Recommendations

CFR requirements

Particular focus on repair station maintenance functions and authority as reflected on the certificate and operations specifications.

- 14 CFR Part 1
- 14 CFR Part 21 (general overview)
- 14 CFR Part 43
- 14 CFR Part 45
- 14 CFR Part 91/121/129/135 (maintenance regulations as appropriate)
- 14 CFR Part 145

Company manuals, policies, procedures, and practices

- Repair Station Ratings (defines Ratings)
- Repair Station Operation Specifications and Limitations (defines limitation of the ratings)
- Repair Station Capabilities (defines products to be worked on)

Risk Assessment / Change Management

AIRCRAFT ELECTRONICS ASSOCIATION

Safety Admin Safety Policy Safety Risk Management Safety Assurance Safety Promotion

SMS Home
Risk Management
Scope of Hazards
Risk Assessment
Safety/Incident/Employee Reporting
What Should Be Reported?

S
M
S

Safety Management

1 2 3 Done

Risk Assessment

Proactively identifying hazards in an operation requires knowledge of the system and analysis of tasks. However, this does not require analysis of every single task performed in a company. Instead, we will focus on the tasks most likely to lead to harm or danger.

All reported hazards or safety concerns will be assessed by the SMS Coordinator regardless of determined Risk Level.

What is your business doing?
Select one

Give a brief description of the change: 2

Create Assessment

Developer note:
User would fill out this form and then click create assessment. System creates new report and goes to step 2 the Physical Consequences page.
[Click here to go to Physical Consequences form](#)

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Learning Cultures

- Learning is seen as more than a requirement for initial skills training; rather it is valued as a lifetime process.
- People are encouraged to develop and apply their own skills and knowledge to enhance organizational safety.
- Staff are updated on safety issues by management, and safety reports are fed back to staff so that everyone can learn the pertinent safety lessons.

AEA Participation

- Regular highlights in Avionics News
- Top ten hazards list on AEA/SMS
- Hazard topics for AEA Convention & Regional Conferences



Reporting Cultures

- Managers and operational personnel freely share critical safety information without the threat of punitive action.
- This is frequently referred to as creating a corporate reporting culture.
- Personnel are able to report hazards or safety concerns as they become aware of them, without fear of sanction or embarrassment.

Employee Hazard Identification Report



The screenshot displays the AEA Safety Assurance web application. At the top, a red navigation bar contains links for "Safety Admin", "Safety Policy", "Safety Risk Management", "Safety Assurance", and "Safety Promotion". Below this, a sidebar on the left includes "SMS Home" and "Report Hazard". The main content area is titled "Safety Assurance" and features a "Create Hazard Report" section. This section includes a form with the following fields: "Is this an anonymous report?" (a dropdown menu set to "No"), "Department:" (a dropdown menu), "Department Other:" (a text input field with a note: "(Dev note: only show if Other is chosen.)"), "Location of hazard:" (a text input field), "Describe the hazard (observed or performed activity, condition, etc.):" (a large text area), "What could be the outcome or consequence of the hazard?" (a text area), "What factors contribute to the hazard?" (a text area), "How often is this hazard likely to happen?" (a text input field), and "How severe are the possible consequences of this hazard?" (a text input field). The background of the interface features a collage of images, including a "CAUTION" sign and a person talking on a mobile phone. On the left side of the interface, there are three red-outlined boxes containing the letters "S", "M", and "S" stacked vertically.

Just Cultures

- While a non-punitive environment is fundamental for a good reporting culture, the workforce must know and agree on what is acceptable and what is unacceptable behavior.
- Negligence or deliberate violations must not be tolerated by management (even in a non-punitive environment).
- A just culture recognizes that, in certain circumstances, there may be a need for punitive action and attempts to define the line between acceptable and unacceptable actions or activities.

SMS Dashboard



AIRCRAFT ELECTRONICS
ASSOCIATION

Safety AdminSafety PolicySafety Risk ManagementSafety AssuranceSafety Promotion

Home
Safety Policy
Safety Risk Management
Safety Assurance
Safety Promotion

SMS

SMS Dashboard

New Hazard Identification(s)

ID	Topic(what are you doing)	Date	Submitter	Status
1234	Test copy	12/12/2012	Aaron Ward	Needs Assessment

New Hazard Report(s)

ID	Topic	Date	Submitter	Status
1234	Test copy	12/12/2012	anonymous	open

Open Hazard Identification(s)

ID	Topic(what are you doing)	Open Date	Reviewer	Status	Review Date
1234	Test copy	12/12/2012	Aaron Ward	waiting for review	7/1/2012

Open Hazard Report(s)

ID	Topic	Open Date	Reviewer	Status	Review Date
1234	Test copy	12/12/2012	Aaron Ward	waiting for review	7/1/2012

Closed Hazard Identification(s)

ID	Topic(what are you doing)	Date Opened	Submitter	Date Closed
1234	Test copy	12/12/2011	Aaron Ward	2/15/2012

Closed Hazard Report(s)

ID	Topic	Date Opened	Submitter	Date Closed
1234	Test copy	12/12/2010	anonymous	1/5/2011

Developer note:
The dashboard is meant to give the user a easy overview of what HDs and HReps are new, in process, and closed. The user will be able to click on the ID to bring up the report to process, review, or read.

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Continuous Improvement

- AEA Safety Committee
 - Quarterly meetings
 - Review hazard classifications
 - Develop top 12 list

Continuous Improvement

- Continuous Improvements in the program
 - Revised SMS Manual
 - Updated to reflect changes in the program
 - Large facility planning
 - Standardized Taxonomy

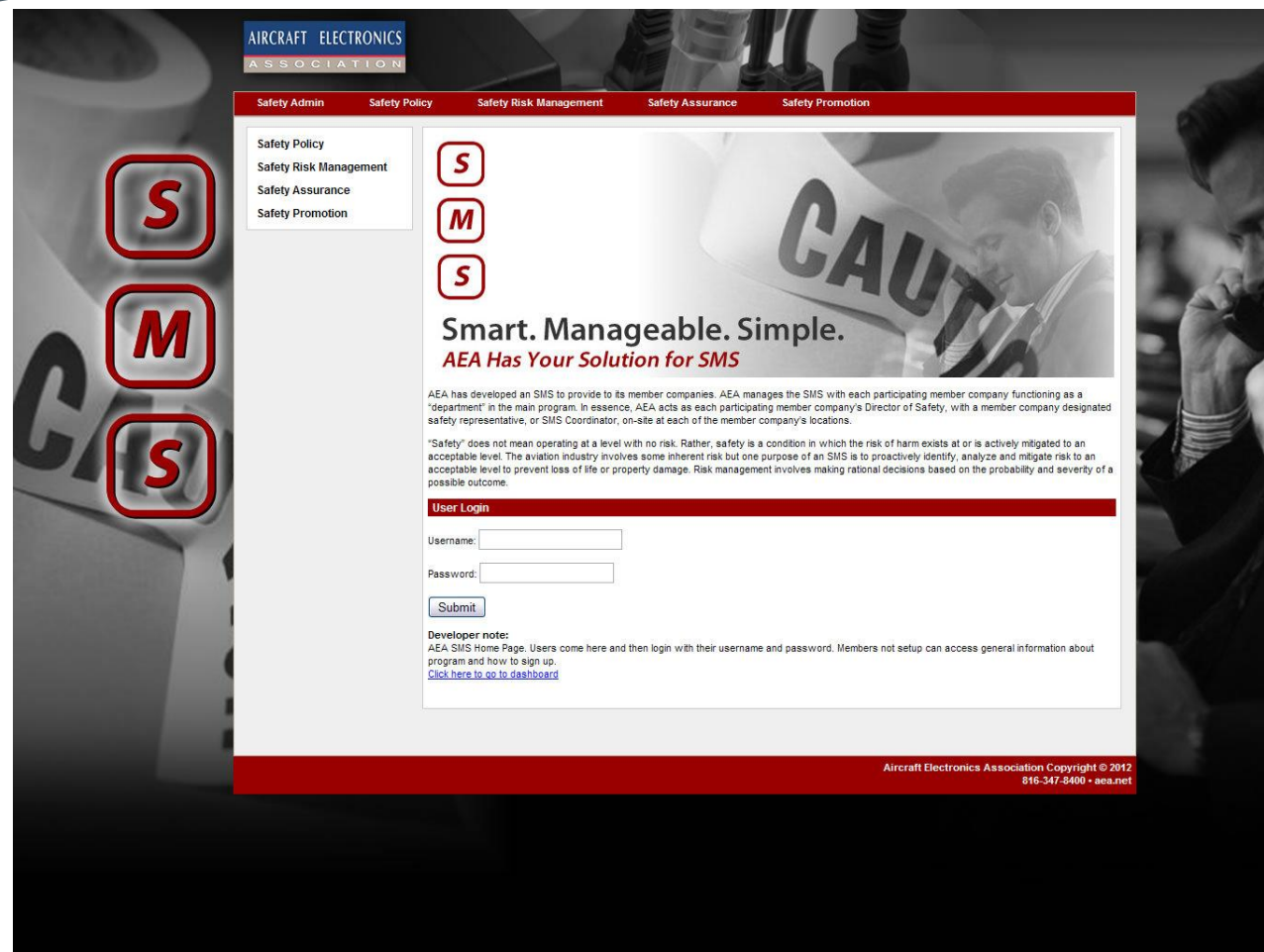
Safety Management Systems

AEA's Solution: Helping the Membership

Smart
Manageable
Simple



A Complete Program: NOT a Template



Safety Policy



The screenshot displays the Aircraft Electronics Association (AEA) website's Safety Policy page. The page has a dark blue header with the AEA logo and navigation links: Safety Admin, Safety Policy, Safety Risk Management, Safety Assurance, and Safety Promotion. A left sidebar contains a vertical stack of three red squares with white letters 'S', 'M', and 'S' (representing SMS), and a list of links: SMS Home, Policy, Commitment, Elements of SMS, and Glossary. The main content area features a large image of a person looking at a 'CAUTION' sign, with the title 'Safety Policy' and a sub-section 'Elements of SMS'. Below this, a list of 13 elements is provided, followed by a 'Developer note' section. The footer contains the copyright notice: Aircraft Electronics Association Copyright © 2012.

AIRCRAFT ELECTRONICS ASSOCIATION

Safety Admin | **Safety Policy** | Safety Risk Management | Safety Assurance | Safety Promotion

SMS Home
Policy
Commitment
Elements of SMS
Glossary

S
M
S

Safety Policy

Elements of SMS

The important elements of the SMS are:

1. Management commitment to the company Safety Management System
2. The SMS Coordinator is the manager of the SMS and either reports directly to the Accountable Manager or is the Accountable Manager
3. Encouragement of a positive safety culture
4. Establishment of a safety management structure
5. Hazard identification and risk management
6. Safety audits and assessment of quality and compliance
7. Accident and incident reporting and investigation
8. Emergency Response Program Management
9. Documentation and Recordkeeping
10. Immunity-based reporting systems
11. The exchange of valuable "Lessons Learned" with others in the industry
12. Safety training integration into the organization's training syllabi
13. Human factors training for all personnel and Regular evaluation and ongoing fine-tuning of the Safety Management System.

Developer note:

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Safety Policy



AIRCRAFT ELECTRONICS
ASSOCIATION

Safety AdminSafety PolicySafety Risk ManagementSafety AssuranceSafety Promotion

SMS Home
Policy
Commitment
Elements of SMS
Glossary

S
M
S

S
M
S

Safety Policy

Commitment

[Company name] has decided to participate in the Aircraft Electronics Association's Safety Management System. As an organization, we have committed to implementing and maintaining this SMS. We do this as part of our efforts to continuously improve the safety of our maintenance operations and comply with regulatory requirements. This SMS will establish a proactive means of controlling risk, thereby improving safety.

All employees are expected to become familiar with the policies and procedures set forth in this manual and to actively participate in this program. Accordingly, each employee, regardless of position, title or duties, will receive initial and recurrent training in the basic concepts of SMS. All employees are expected to conduct their duties with safety, quality, and regulatory compliance as main priorities.

I personally commit [company name] to this process and assure you resources will be assigned to implementing and maintaining this SMS as appropriate. Further, I encourage you to report errors or potential safety issues without fear of reprisal. Unless an action is the result of neglect, intentional conduct, or use of drugs or alcohol, no report will result in disciplinary action, termination, or other negative actions.

Sincerely,

Accountable Executive

Developer note:

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Employee Hazard Identification Report



The screenshot displays the Aircraft Electronics Association (AEA) Safety Assurance web application. The header includes the AEA logo and navigation links: Safety Admin, Safety Policy, Safety Risk Management, Safety Assurance, and Safety Promotion. The main content area is titled "Safety Assurance" and features a "Create Hazard Report" section. The form includes the following fields:

- Is this an anonymous report? ☐ No ☐ Yes
- Department:
- Department Other: (Dev note: only show if Other is chosen.)
- Location of hazard:
- Describe the hazard (observed or performed activity, condition, etc.):
- What could be the outcome or consequence of the hazard?:
- What factors contribute to the hazard?:
- How often is this hazard likely to happen?:
- How severe are the possible consequences of this hazard?:

The background of the application shows a person working on an aircraft engine, with a "CAUTION" sign visible.

Risk Assessment / Change Management



The screenshot displays the AEA Safety Management System (SMS) interface. At the top, the 'AIRCRAFT ELECTRONICS ASSOCIATION' logo is visible. Below it, a red navigation bar contains links: 'Safety Admin', 'Safety Policy', 'Safety Risk Management', 'Safety Assurance', and 'Safety Promotion'. The left sidebar lists menu items: 'SMS Home', 'Risk Management', 'Scope of Hazards', 'Risk Assessment', 'Safety/Incident/Employee Reporting', and 'What Should Be Reported?'. The main content area is titled 'Safety Management' and features a progress indicator with steps 1, 2, 3, and 'Done'. Step 1 is currently active. The 'Risk Assessment' section explains that hazards are identified proactively based on system knowledge and task analysis. It includes a dropdown menu for 'What is your business doing?' and a text area for 'Give a brief description of the change:'. A 'Create Assessment' button is located below the text area. A 'Developer note' at the bottom states that users should fill out the form and click 'create assessment' to generate a new report and proceed to step 2. The footer contains the copyright information: 'Aircraft Electronics Association Copyright © 2012 816-347-5400 • aea.net'.

AIRCRAFT ELECTRONICS ASSOCIATION

Safety Admin Safety Policy Safety Risk Management Safety Assurance Safety Promotion

SMS Home
Risk Management
Scope of Hazards
Risk Assessment
Safety/Incident/Employee Reporting
What Should Be Reported?

S
M
S

CAUTION

Safety Management

1 2 3 Done

Risk Assessment

Proactively identifying hazards in an operation requires knowledge of the system and analysis of tasks. However, this does not require analysis of every single task performed in a company. Instead, we will focus on the tasks most likely to lead to harm or danger.

All reported hazards or safety concerns will be assessed by the SMS Coordinator regardless of determined Risk Level.

What is your business doing?
Select one

Give a brief description of the change: 2

Create Assessment

Developer note:
User would fill out this form and then click create assessment. System creates new report and goes to step 2 the Physical Consequences page.
[Click here to go to Physical Consequences form](#)

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SMS Dashboard



AIRCRAFT ELECTRONICS
ASSOCIATION

Safety AdminSafety PolicySafety Risk ManagementSafety AssuranceSafety Promotion

Home
Safety Policy
Safety Risk Management
Safety Assurance
Safety Promotion

SMS

SMS Dashboard

New Hazard Identification(s)

ID	Topic(what are you doing)	Date	Submitter	Status
1234	Test copy	12/12/2012	Aaron Ward	Needs Assessment

New Hazard Report(s)

ID	Topic	Date	Submitter	Status
1234	Test copy	12/12/2012	anonymous	open

Open Hazard Identification(s)

ID	Topic(what are you doing)	Open Date	Reviewer	Status	Review Date
1234	Test copy	12/12/2012	Aaron Ward	waiting for review	7/1/2012

Open Hazard Report(s)

ID	Topic	Open Date	Reviewer	Status	Review Date
1234	Test copy	12/12/2012	Aaron Ward	waiting for review	7/1/2012

Closed Hazard Identification(s)

ID	Topic(what are you doing)	Date Opened	Submitter	Date Closed
1234	Test copy	12/12/2011	Aaron Ward	2/15/2012

Closed Hazard Report(s)

ID	Topic	Date Opened	Submitter	Date Closed
1234	Test copy	12/12/2010	anonymous	1/5/2011

Developer note:
The dashboard is meant to give the user an easy overview of what HIDs and HReps are new, in process, and closed. The user will be able to click on the ID to bring up the report to process, review, or read.

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Smart – Manageable - Simple

- You run your business – we give you the tools
 - You make the decisions – not us.
- Complete program – NOT a template
 - AEA Program -- Coordinated with NAAs
- Employee participation
 - Every employee is enrolled and has access to the program



Smart – Manageable - Simple

- Intuitive
- All the tools in one place
 - www.aea.net

Full Facility Audit

- Quality and Safety
 - Performed progressively throughout the year
 - 1/12 of the audit each month
 - Simple – configurable audits
 - including:
 - Repair station quality audit
 - Industrial safety elements
 - Risk mitigation solutions

Smart – Manageable - Simple

■ Training

- We learn from each other
- Training at the AEA Regional Meetings
- Generic on-line safety/SMS training
- Facility specific training templates

Emergency Response Manual

- The Emergency Response Manual serves as guidance for all personnel. It provides a clear, timely, and ordered response to an emergency.
- Often required by local government.

Emergency Response Manual

- Media and PR
 - Planning Communications

S

M

S

Smart. Manageable. Simple.

How the System Works



SMS Sign-In



AIRCRAFT ELECTRONICS ASSOCIATION

- Safety Policy
- Safety Promotion
- Safety Risk Management
- Safety Assurance

SMS

Smart. Manageable. Simple.
The AEA Has Your Solution for SMS

The Aircraft Electronics Association has developed a safety management systems program for its member companies. The AEA manages the SMS with each participating member company functioning as a "department" in the main program. In essence, the AEA acts as each participating member company's director of safety, with a member company designated safety representative, or SMS coordinator, on-site at each of the member company's locations.

"Safety" does not mean operating at a level with no risk. Rather, safety is a condition in which the risk of harm exists at or is actively mitigated to an acceptable level. The aviation industry involves some inherent risk, but one purpose of an SMS is to proactively identify, analyze and mitigate risk to an acceptable level to prevent loss of life or property damage. Risk management involves making rational decisions based on the probability and severity of a possible outcome.

SMS Enrollment Process

1. Attend AEA SMS coordinator training (conducted at AEA Regional Meetings and International Convention & Trade Show).
2. Identify by name and email contact the company accountable executive and SMS coordinator (SMSCO) to the AEA.
3. The AEA will distribute an employee enrollment form. The SMS coordinator will return the completed form with all employee names, unique email addresses, user names and passwords.
4. The accountable executive signs and posts the Safety Management System Website Agreement and notifies the employees of the company's participation in the AEA SMS program.
5. All employees login to the SMS site and complete the Employee Initial Training (EIT) program.
6. The accountable executive signs and posts the Safety Policy Commitment.

Enter your information and click the login button:

Username:

Password:

☐ remember me

[Login](#)

[Forgot Password?](#)

SMS Dashboard



S

M

S

Safety Policy

Safety Promotion

Safety Risk Management

Safety Assurance

AIRCRAFT ELECTRONICS ASSOCIATION

Welcome Ric Pen
Aircraft Electronics Association
Logout

Safety Policy

Safety Promotion

Safety Risk Management

Safety Assurance

User Settings

SMS Dashboard

Employee Hazard Report(s) submitted by Ric Pen

ID	Description	Date	Status
11127	www.....	3/15/2012	Identify
11128	Test GP...	3/15/2012	Identify

Employee Hazard Report(s)

ID	Description	Date	Status
11127	www.....	3/15/2012	Identify
11128	Test GP...	3/15/2012	Identify
11129	sd...	3/15/2012	Corrective Action
11130	sd...	3/24/2012	Evaluate
11131	food stored near chemicals...	3/15/2012	Evaluate
11132	sd...	3/15/2012	Identify
11133	sd...	3/15/2012	Evaluate
11134	frayed power cables...	11/12/2012	Evaluate
11135	floor drain cover is missing...	10/21/2012	Identify
11136	faulty hanger door rail door slides off ...	10/17/2012	Evaluate
11137	sd...	10/17/2012	Identify
11138	fluid on the hanger floor...	10/10/2012	Identify
11139	not well ventilated...	9/29/2012	Evaluate
11140	sd...	9/11/2012	Evaluate
11141	spilled fluid...	9/29/2012	Evaluate
11142	power cord stretched across floor...	1/19/2012	Corrective Action

Incident/Accident Reports

ID	Description	Date	Status
11143	technician tripped and landed hard on their hands ...	10/10/2012	Evaluate
11144	sd...	12/17/2012	Corrective Action
11145	sd...	1/10/2013	Identify
11146	A USOKUP...	3/24/2012	Corrective Action
11147	sd...	3/12/2012	Identify
11148	sd...	3/12/2012	Identify
11149	sd...	3/15/2012	Assess
11150	sd...	3/17/2012	Corrective Action

Audit Finding Reports

ID	Description	Date	Status
11151	sd...	3/17/2012	Documented
11152	sd...	3/12/2012	Corrective Action
11153	insufficient work space and stress for the proper s...	3/7/2012	Evaluate
11154	sd/ceat/d...	3/6/2012	Identify
11155	Just being to see if his works...	3/6/2012	Identify
11156	sd...	3/24/2012	Corrective Action
11157	main door propped open - security module auto stan...	3/19/2012	Documented
11158	sd...	3/11/2012	Corrective Action
11159	sd...	1/14/2012	Evaluate

Safety Policy

AIRCRAFT ELECTRONICS
ASSOCIATION

Policy
Commitment
Elements of SMS
Glossary

Safety Assurance

Archive

Welcome Ric Peri
Aircraft Electronics Association

Logout

Safety PolicySafety PromotionSafety Risk ManagementSafety AssuranceUser Settings

SMS Dashboard

Employee Hazard Report(s) submitted by Ric Peri

ID	Description	Date	Status
11197	vvvvvvvvvvv...	3/15/2013	Identify
11198	Test EP...	3/15/2013	Identify

Safety Promotion

AIRCRAFT ELECTRONICS
ASSOCIATION

Training
Information Distribution
Newsletter

Safety Policy
Safety Promotion
Safety Risk Management
Safety Assurance
Archive

Welcome Ric Peri
Aircraft Electronics Association
Logout

Safety PolicySafety PromotionSafety Risk ManagementSafety AssuranceUser Settings

SMS Dashboard

Employee Hazard Report(s) submitted by Ric Peri

ID	Description	Date	Status
11197	vvvvvvvvvvvv...	3/15/2013	Identify
11198	Test EP...	3/15/2013	Identify

Safety Risk Management

AIRCRAFT ELECTRONICS ASSOCIATION

Welcome Ric Peri
Aircraft Electronics Association
Logout

Safety Policy Safety Promotion Safety Risk Management Safety Assurance User Settings

SMS Dashboard

Employee Hazard Report(s) submitted by Ric Peri

ID	Description	Date	Status
11197	vvvvvvvvvvv...	3/15/2013	Identify
11198	Test EP...	3/15/2013	Identify

Hazard Identification

The screenshot displays the Hazard Identification web application interface. At the top, a red navigation bar contains links for Safety Policy, Safety Promotion, Safety Risk Management, Safety Assurance, and User Settings, with a Logout link in the top right corner. A left sidebar features three large red buttons labeled 'S', 'M', and 'S'. The main content area is titled 'Hazard Identification' and includes a search icon. Below this, there are three sections: 'Employee Hazard Identification Report' with a yellow caution tape icon and a link to 'Create Employee Hazard Identification Report'; 'Incident/Accident' with a first aid kit icon and a link to 'Click here to create Incident/Accident Report'; and 'Organizational Risk Assessment' with a yellow lightning bolt icon and a link to 'Click here to create Risk Assessment'. The footer contains the AEA logo and copyright information: 'Aircraft Electronics Association Copyright © 2013 816-347-3400 - aea.net'.

Logout

Safety Policy Safety Promotion Safety Risk Management Safety Assurance User Settings

Dashboard

Hazard Identification

Employee Hazard Identification Report

 [Create Employee Hazard Identification Report](#)

Incident/Accident

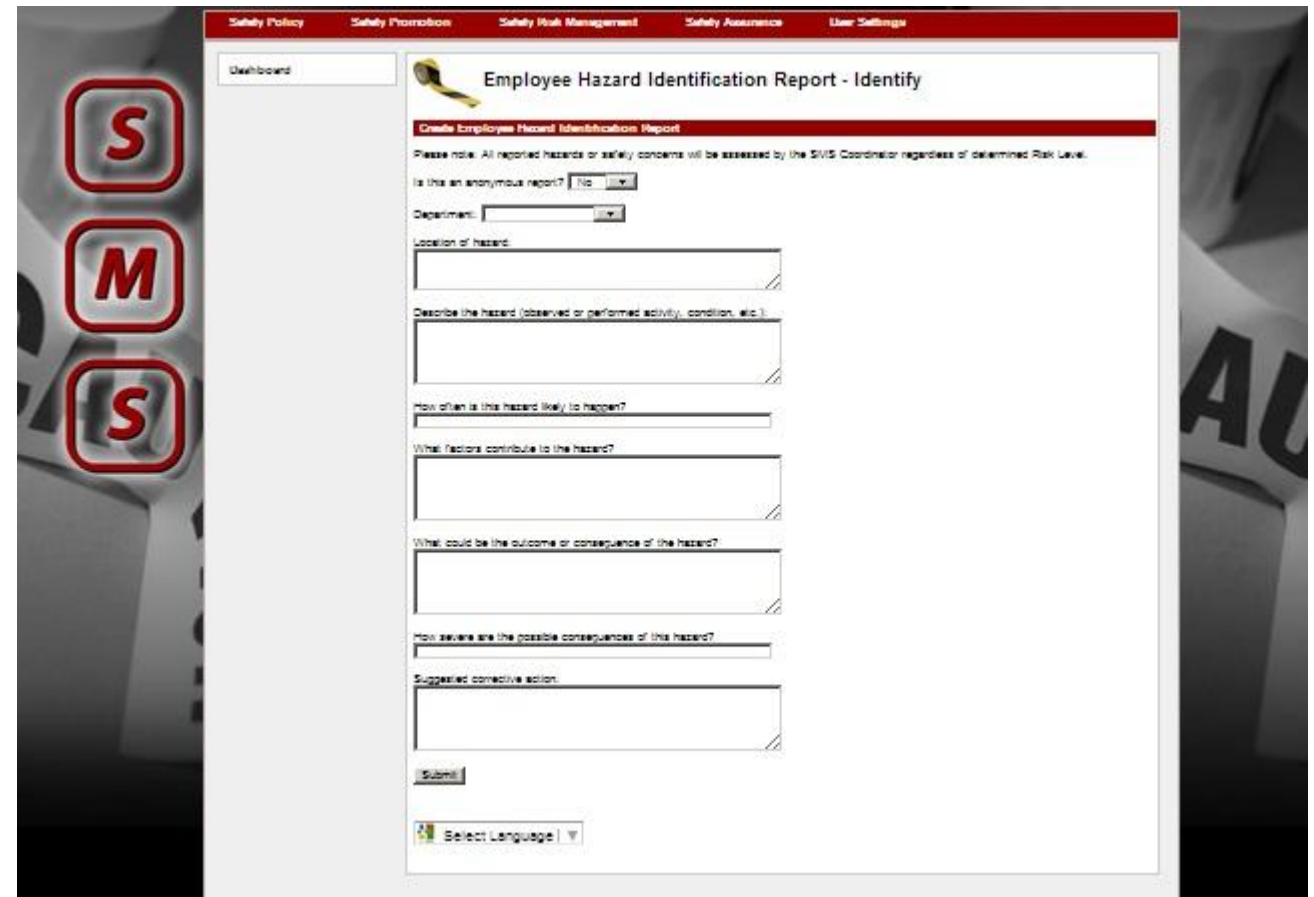
 [Click here to create Incident/Accident Report](#)

Organizational Risk Assessment

 [Click here to create Risk Assessment](#)

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Employee Hazard Identification Report



The screenshot displays a web application interface for creating an Employee Hazard Identification Report. The top navigation bar is red and contains links for Safety Policy, Safety Promotion, Safety Risk Management, Safety Assurance, and User Settings. A sidebar on the left features three red square buttons with white letters: S, M, and S. The main content area is titled "Employee Hazard Identification Report - Identify" and includes a sub-header "Create Employee Hazard Identification Report". A note states: "Please note: All reported hazards or safety concerns will be assessed by the SMS Coordinator regardless of determined Risk Level." The form contains several fields: "Is this an anonymous report?" with a dropdown menu set to "No"; "Department:" with a dropdown menu; "Location of hazard:" with a text input field; "Describe the hazard (observed or performed activity, condition, etc.):" with a text input field; "How often is this hazard likely to happen?" with a text input field; "What factors contribute to the hazard?" with a text input field; "What could be the outcome or consequence of the hazard?" with a text input field; "How severe are the possible consequences of this hazard?" with a text input field; and "Suggested corrective action:" with a text input field. A "Submit" button is located at the bottom of the form. At the very bottom, there is a "Select Language" dropdown menu.

Employee Hazard Identification Report

[Safety Policy](#) [Safety Promotion](#) [Safety Risk Management](#) [Safety Assurance](#) [User Settings](#)

Dashboard



Employee Hazard Identification Report - Identify

Create Employee Hazard Identification Report

Please note: All reported hazards or safety concerns will be assessed by the SMS Coordinator regardless of determined Risk Level.

Is this an anonymous report?

Department:

Location of hazard:

Describe the hazard (observed or performed activity, condition, etc.):

How often is this hazard likely to happen?

Employee Hazard Identification Report



[Safety Policy](#) [Safety Promotion](#) [Safety Risk Management](#) [Safety Assurance](#) [User Settings](#)

Dashboard

 **Employee Hazard Identification Report - Identify**

Create Employee Hazard Identification Report

Please note: All reported hazards or safety concerns will be assessed by the SMS Coordinator regardless of determined Risk Level.

Is this an anonymous report?

Department:

Location of hazard:

Describe the hazard (observed or performed activity, condition, etc.):

Employee Hazard Identification Report

How often is this hazard likely to happen?

What factors contribute to the hazard?

What could be the outcome or consequence of the hazard?

How severe are the possible consequences of this hazard?

Suggested corrective action:

Submit

Employee Hazard Identification Report

What could be the outcome or consequence of the hazard?

How severe are the possible consequences of this hazard?

Suggested corrective action:

Submit

Select Language ▼

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Assessment

Logout

Safety Policy Safety Promotion Safety Risk Management Safety Assurance User Settings

Dashboard

Employee Hazard Identification Report - Assessment

Employee Hazard Identification Report Details - # 11196

#	11196
Reported On:	3/15/2013
Reported By:	Ric Peri
Department:	Maintenance
Location:	Test EP
Description:	Test EP
Frequency:	Test EP
Factors:	Test EP
Outcome:	Test EP
Severity:	Test EP

Assessment

Outcome: Test EP
Severity: Test EP
Suggested Resolution: Test EP

Investigation

Verify Employee Hazard Identification Report description and enter additional information from investigation:

Verify and identify contributing factors:

Safety Analysis

How often is this hazard likely to happen?

☐ Frequent ☒ Rare

Assessment

Safety Analysis

How often is this hazard likely to happen?

☐ Frequent ?

☐ Occasional ?

☐ Remote ?

☐ Improbable ?

☐ Extremely Improbable ?

How severe are the possible consequences of this hazard?

☐ Catastrophic ?

☐ Hazardous ?

☐ Major ?

☐ Minor ?

☐ Negligible ?

What were the root cause(s) that directly resulted in the hazard?

Immediate Action

Risk Probability & Severity Table

Risk Probability	Risk Severity				
	Catastrophic A	Hazardous B	Major C	Minor D	Negligible E
Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E
Extremely Improbable 1	1A	1B	1C	1D	1E

Assessment

Immediate Action

(If Required)

Save Assessment Progress

Save Assessment & Proceed to Corrective Action

Corrective Action

Corrective Action

Describe appropriate corrective action plan:

Accountable person assigned:

Date of implementation: (MM/DD/YYYY)

Date for review: (MM/DD/YYYY)

Person assigned for review:

Organizational Risk Assessment

AIRCRAFT ELECTRONICS
ASSOCIATION

Welcome Ric Peri
Aircraft Electronics Association
Logout

Safety PolicySafety PromotionSafety Risk ManagementSafety AssuranceUser Settings

Dashboard



Organizational Risk Assessment - Create Report

Proactively identifying hazards in an organization requires knowledge of the system and analysis of tasks. However, this does not require analysis of every single task performed in a company. Instead, we will focus on the tasks most likely to lead to harm or danger.

When a major change in operations, equipment or services is anticipated, this assessment process will be used and will include hazard identification and risk management processes.

Reason for Risk Assessment?

Select one

Give a brief description: ?

Organizational Risk Assessment

Physical Consequences

Identify one specific hazard related to the generic hazard shown above: ?

Identify one physical consequence related to the specific hazard entered above: ?

How often is this specific hazard likely to happen?

- ☐ Frequent ?
- ☐ Occasional ?
- ☐ Remote ?
- ☐ Improbable ?
- ☐ Extremely Improbable ?

How severe is the specific physical consequence?

- ☐ Catastrophic ?
- ☐ Hazardous ?
- ☐ Major ?
- ☐ Minor ?
- ☐ Negligible ?

Organizational Risk Assessment

☐ Minor 

☐ Negligible 

Mitigate Physical Consequence (If Required or needed)

[Save & Add Another Physical Hazard](#)

[Save & Continue to Human Consequences](#)

[Continue to Human Consequences](#)

Organizational Risk Assessment

Human Consequences

Identify one specific hazard related to the generic hazard shown above: ?

Identify one human consequence related to the specific hazard entered above: ?

How often is this specific hazard likely to happen?

- ☐ Frequent ?
- ☐ Occasional ?
- ☐ Remote ?
- ☐ Improbable ?
- ☐ Extremely Improbable ?

How severe is the human consequence?

- ☐ Catastrophic ?
- ☐ Hazardous ?
- ☐ Major ?
- ☐ Minor ?
- ☐ Negligible ?

Organizational Risk Assessment

☐ Major ?
☐ Minor ?
☐ Negligible ?

Mitigate Human Consequence (If Required or needed)

Save & Add Another Human Hazard

Save & Continue to Summary

Continue to Summary

Organizational Risk Assessment



Organizational Risk Assessment - Summary

Summary:

Hazard ID: 11201
Started On: 3/23/2013
Created By: 1114
Reason: Major change in the facility
Generic Hazard: mmmmmmmmmmm

List of Physical Consequences

Consequence	Mitigation	Acceptability	
mmmmmmmmmmmmmmmmmmmm		2D	Edit Delete

Add Physical Consequence

Add Human Consequence

Safety Assurance

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Safety Policy Safety Promotion Safety Risk Management **Safety Assurance User Settings**

Safety Policy
Safety Promotion
Safety Risk Management
Safety Assurance

Archive

 **SMS Dashboard**

- Dashboard
- Auditing Finding
- Internal Evaluation Program

Employee Hazard Report(s) submitted by Ric Peri

ID	Description	Date	Status
11197	vvvvvvvvvvvv...	3/15/2013	Identify
11198	Test EP...	3/15/2013	Identify

Dashboard

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Safety Policy

Safety Promotion

Safety Risk Management

Safety Assurance

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Safety Risk Management

Safety Assurance

User Settings

 SMS Dashboard

Employee Hazard Report(s) submitted by Ric Pen

ID	Description	Date	Status
1127	xxxxxxxxxxxxxx	2/15/2012	Identify
1128	Test SP...	2/15/2012	Identify

Employee Hazard Report(s)

ID	Description	Date	Status
1127	xxxxxxxxxxxxxx	2/15/2012	Identify
1128	Test SP...	2/15/2012	Identify
1129	sd...	2/15/2012	Corrective Action
1130	sd...	2/24/2012	Evaluate
1131	Food stored near chemicals...	2/15/2012	Evaluate
1132	sd sd sd sd...	2/1/2012	Identify
1133	sd in floor...	1/21/2012	Evaluate
1134	frayed power cables...	1/11/2012	Evaluate
1135	floor drain cover is missing...	10/21/2012	Identify
1136	faucy hanger door rel. door slides off...	10/11/2012	Evaluate
1137	sd...	10/11/2012	Identify
1138	fluid on the hanger floor...	10/10/2012	Identify
1139	not well ventilated...	9/25/2012	Evaluate
1140	oil floor...	9/11/2012	Evaluate
1141	spilled fuel...	9/29/2012	Evaluate
1142	power cord stretched across floor...	7/19/2012	Corrective Action

Incident/Accident Reports

ID	Description	Date	Status
21118	technician tripped and landed hard on their hands...	12/10/2012	Evaluate
21119	sd...	12/11/2012	Corrective Action
21120	sd...	1/10/2013	Identify
21121	A LSC/R...	2/24/2013	Corrective Action
21122	sd...	2/13/2013	Identify
21123	sd...	3/12/2013	Identify
21124	sd...	3/15/2013	Assess
21125	sd...	3/11/2013	Corrective Action

Audit Finding Reports

ID	Description	Date	Status
21112	sd...	2/17/2013	Documented
21113	sd...	2/12/2013	Corrective Action
21114	insufficient work space and access for the proper a...	2/7/2013	Evaluate
21115	sd/Case 2...	2/5/2013	Identify
21116	Just looking to see if this works...	2/5/2013	Identify
21117	sd...	2/24/2013	Corrective Action
21118	handcuff dropped open - security module audit stan...	2/17/2013	Documented
21119	excesses...	2/11/2013	Corrective Action
21120	audit finding...	1/14/2013	Evaluate

Internal Evaluation Program



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Dashboard Internal Evaluation Program

Internal Audit

Month	Audit	Link
January	Module 1: Management	PDF
February	Module 2: Recordkeeping & Manual Management	PDF
March	Module 2: Repair Station Manual / Maintenance Control	PDF
April	Module 4: Quality Control & Assurance	PDF
May	Module 5: Process Evaluation	PDF
June	Module 6: Safety Management System	PDF
July	Module 7: Stores / Shelf Life Inspection Program	PDF
August	Module 8: Occupational Safety & Health and Hazardous Materials/Dangerous Goods	PDF
September	Module 9: Personnel	PDF
October	Module 10: Training	PDF
November	Module 11: Security	PDF
December	Module 12: Human Resources	PDF

Top Hazards

Number	Hazard
1	Personnel using top step of ladders, sitting or standing. Lack of understanding in how to work at various heights.
2	Waste fluid handling. Disposal practices not being followed.
3	Improper use of torque wrench and voltmeter, lack of understanding on proper usage.
4	Incorrect solder practices, audio ground rail found with untinned wires, broken from rail.
5	Aircraft parked such that wings overhang designated walkways which are outlined in yellow around perimeter of hangar.
6	AC power strip, wall plug melted and burnt. Power strip still in use, not identified as damaged.
7	Zip ties used to suspend lighting in an aircraft broke when the plug was pulled. The zip tie was undersized for the task.
8	Access stand positioned at aircraft entrance moved where leg fell ended up on an extension cord. Caused employee to stumble and use the aft door as a brace to prevent falling. The aft door contacted the rear baggage door resulting in minor damage.
9	Outlined tool bag on floor, multiple jobs ongoing.
10	Cups with water placed on work bench.
11	Not following procedures regarding options, modifications, and service bulletin level in operating system updates.

Audit

Internal Audit

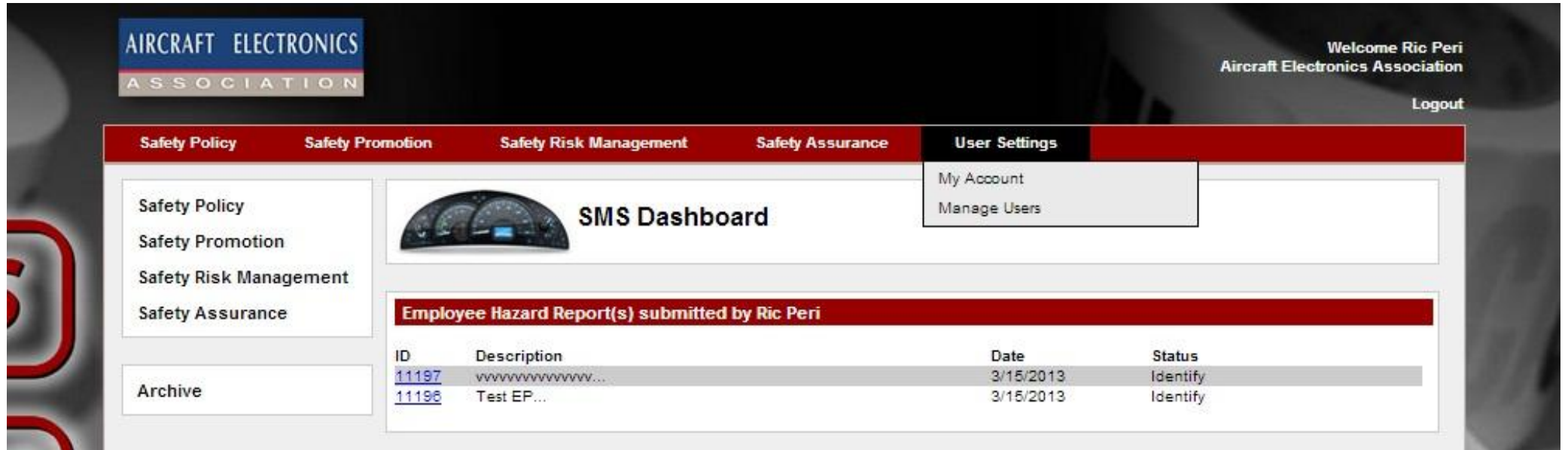
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7	Zip ties used to suspend lighting in an aircraft broke when the plug was pulled. The zip tie was undersized for the task.
8	Access stand positioned at aircraft entrance moved where leg feet ended up on an extension cord. Caused employee to stumble and use the aft door as a brace to prevent falling. The aft door contacted the rear baggage door resulting in minor damage.
9	Cluttered tool bag on floor, multiple jobs ongoing.
10	Cups with water placed on work bench.
11	Not following procedures regarding "options, modifications, and service bulletin level" in operating system updates.
12	Airplane seats not being properly installed following avionics maintenance.

User Settings



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Safety Policy **Safety Promotion** **Safety Risk Management** **Safety Assurance** **User Settings**

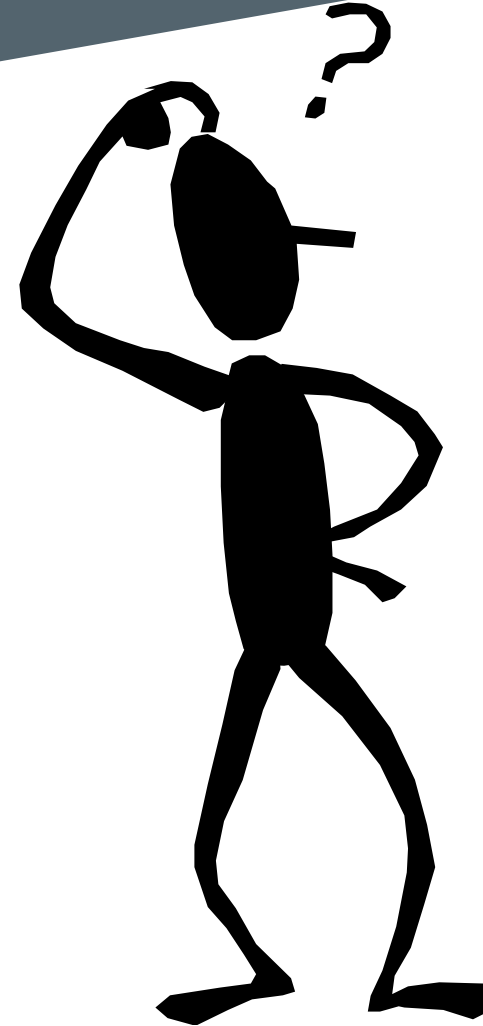
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 **SMS Dashboard**

Employee Hazard Report(s) submitted by Ric Peri

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Questions?



14 CFR Part 5

Regulatory Foundation for SMS (14 CFR Part 5)

- Establishes SMS requirements for aviation service providers
- Defines minimum SMS components and structure
- Basis for FAA SMS Voluntary Program (SMSVP)
- Referenced in FAA/EASA MAG and InFO 24007

Structure of 14 CFR Part 5

- Subpart A – General
- Subpart B – Safety Policy
- Subpart C – Safety Risk Management
- Subpart D – Safety Assurance
- Subpart E – Safety Promotion
- Subpart F – SMS Documentation

ICAO SMS Framework vs Part 5

ICAO	Part 5
Safety Policy	Subpart B
Safety Risk Management	Subpart C
Safety Assurance	Subpart D
Safety Promotion	Subpart E

Part 5 Meets Part 145

- Part 145 defines WHAT must be done
- Part 5 defines HOW safety is managed
- 145.211 Quality System ↔ Part 5 SMS
- 145.163 Training ↔ 5.91 Competency
- 145.151 Personnel ↔ 5.23 Accountability
- Manuals ↔ Subpart F SMS Documentation

How AEA Aligns with Part 5

- Framework aligned to Subparts A–F
- Integrates with Part 145 manuals and procedures
- Supports SMSVP compliance
- Enables EASA acceptance



Regulatory Bottom Line

- Part 5 = SMS framework
- ICAO = international standard
- SMSVP = implementation pathway
- MAG/InFO = enforcement driver

Part 5

Getting Started

Safety Management System

- *Getting Started*

Step 1

■ SMS Coordinator Training

Step 2

Leadership Commitment



Step 2: Leadership Commitment

- Senior leadership must understand the value of SMS in establishing and achieving the organizational goals.

Step 2: Leadership Commitment

- Understands the Value of SMS
- Places SMS on par with other “management Systems”
 - Financial management
 - Production management

Step 2: Leadership Commitment

- Identify and assign SMS Program Manager

Step 3

Leadership Promotion

Step 3: Leadership Promotion

- Before SMS can become effective, senior leadership must promote their commitment to this program.

Step 3: Top-Down Promotion

- Senior leadership must meet with each level – independently - on the organizational chart beginning at the top.

“this is how we do things here”

Step 4

- Communicate with AEA requesting enrollment:
 - Email to Aaron Ward (aaronw@aea.net)
 - Copy Pam Meyer (pamm@aea.net)

Step 5

- Provide AEA (Aaron Ward) a list of all employees so we can be sure we have an account setup for each of them.
 - Include:
 - First Name,
 - Last Name, and
 - email address.



Step 6

■ Employee Training

Step 7

- Establish safety goals and objectives based on local needs and history.

Step 8: Implementation

Begin using:

Safety Promotion

Communication

Outreach

Training



Step 8: Implementation

Begin using:

- Safety Risk Management

- Employee Hazard Reports

- Incident/Accident Reports

- Organizational Risk Assessment



Step 8: Implementation

Begin using:

Safety Assurance

Audit Finding Report

Internal Evaluation Program

Step 9: Evaluation

Is everyone using all of the pieces of SMS?

From the top of the organization chart through EACH level?

Step 9: Evaluation

Are you performing a root cause analysis of each audit finding – regardless of source?

Step 9: Evaluation

Are you performing an internal audit?

Progressively, quarterly or annually?

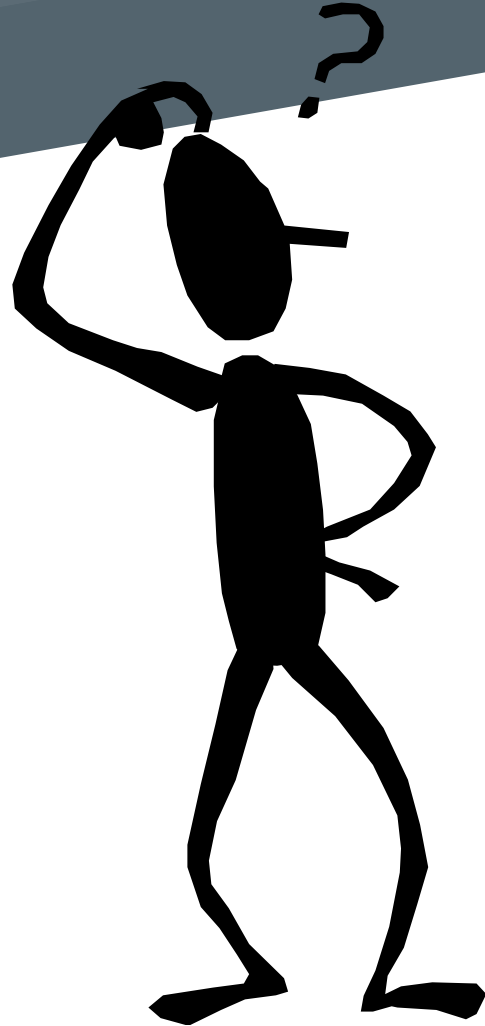
Step 10: Update Goals

- Annually, review and update the organizational safety goals and objectives.

Step 11: Completion

While SMS is operational at start-up, it takes completion of the implementation, evaluation, and internal audit before one can claim to be fully operational.

Questions?



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